

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.29988	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	75.9229	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1579.2179	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	13.06	n/a	131.78534	1	0	0	0.940735	15.39
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	133.2573	1	0	0	0.974807	14.15
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.273435	0.273435	139.37918	0	0	0	0.0	0.0

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.29988.
- Accepted epoch count: 0.
- Fallback epoch count: 0.

- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 75.9229.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1579.2179.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:

- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 8.958333.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 13.06.
- Held-out runtime ms: 131.78534.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 1.616667.
- Worst held-out instances:
- X-n275-k28: feasible=False, penalized gap=15.7, errors=["NameError: name 'isinstance' is not defined", 'missing 274 customers']
- X-n247-k50: feasible=False, penalized gap=14.3, errors=["NameError: name 'isinstance' is not defined", 'missing 246 customers']
- X-n223-k34: feasible=False, penalized gap=13.1, errors=["NameError: name 'isinstance' is not defined", 'missing 222 customers']
- X-n190-k8: feasible=False, penalized gap=11.45, errors=["NameError: name 'isinstance' is not defined", 'missing 189 customers']
- X-n176-k26: feasible=False, penalized gap=10.75, errors=["NameError: name 'isinstance' is not defined", 'missing 175 customers']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 133.2573.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.

- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 139.37918.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary

Aspect	Summary
Direct Synthesis	direct_generate_plus_one_repair had 0% feasibility on train and heldout with very poor penalty gaps (8.96 train, 13.06 heldout), runtime ~132 ms, and highest code novelty (0.94) and complexity (15.39). Not competitive.
Budget-Matched Independent Search	budget_matched_no_replay also showed 0% feasibility (train & heldout), penalized gap 3.0, runtime ~133 ms, high novelty (0.97) and complexity (14.15). Did not yield feasible solutions.
Replay-Aware Iterative Search	replay_solver_evolution had 100% feasibility on train and heldout, penalized gap 0.273, runtime ~139 ms, zero novelty and complexity, indicating reliance on baseline-like solutions but feasible.

Aspect	Summary
Fixed-Baseline Context	Baselines clark_wright_savings and nearest_neighbor_constructive both had 100% feasibility; Clarke-Wright had the lowest heldout gap (0.074) and moderate runtime (~76 ms), nearest neighbor gap was higher (0.273) but faster (~42 ms). Regret insertion local search was slow (~1579 ms) and had worse gap (0.466).
Feasibility	Only fixed baselines and replay condition achieved 100% feasibility on held-out data. Learned direct and budget-matched conditions failed feasibility completely.
Objective Gap	Best gap was from clark_wright_savings baseline (0.074), closely followed by replay search (0.273). Learned/no-replay and direct methods had large or undefined gaps due to infeasibility.
Runtime	Baselines ran quickly (42-76 ms), replay search roughly double nearest neighbor (~139 ms), regret insertion local much slower (~1.58 s). Learned attempts comparable to replay (~130 ms) but infeasible.
Robustness	Baselines showed low robustness dispersion (0.0027-0.0077); replay search matched nearest neighbor dispersion; learned conditions exhibited higher dispersion or zero feasibility, indicating instability.
Best Condition	phase9_closeout_baseline_clark_wright_savings is best heldout condition by gap and stability.
Replay vs. Budget-Matched No-Replay	Replay (replay_solver_evolution) matched feasibility and robustness of baselines and dramatically outperformed budget-matched no-replay control which had 0% feasibility and higher gaps. Replay clearly improved performance over no-replay.
Did any learned method clearly beat fixed baselines?	No. Learned methods either failed feasibility or had much larger gaps than fixed baselines. Baselines remain superior.

Conservative Conclusion

The fixed baselines, particularly Clarke-Wright Savings, achieved consistent 100% feasibility on held-out CVRP instances with the best objective gaps and efficient runtimes. Learned direct synthesis and budget-matched independent search conditions failed to produce feasible solutions in this closeout suite. Replay-aware iterative search maintained feasibility and robustness comparable to fixed baselines and substantially outperformed the budget-matched no-replay condition, indicating that replay improves solution quality and stability. However, no learned condition in this phase outperformed the fixed-baseline context in terms of feasibility or objective gap.